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Section : C4

Assignment : 03

Question 01 : (15.2 Q 23)

$$\int_0^1 \int_0^{y^2} 3y^3 e^{xy} dx dy$$

$$0 \leq x \leq y^2$$

$$0 \leq y \leq 1$$

$$\int_0^1 [3y^3 e^{xy}]_0^{y^2} dy$$

$$\int_0^1 (3y^2 e^{y^3} - 3y^2) dy$$

$$[e^{y^3} - y^3]_0^1 = \boxed{e - 2}$$

Question 02 (15.2 Q 49)

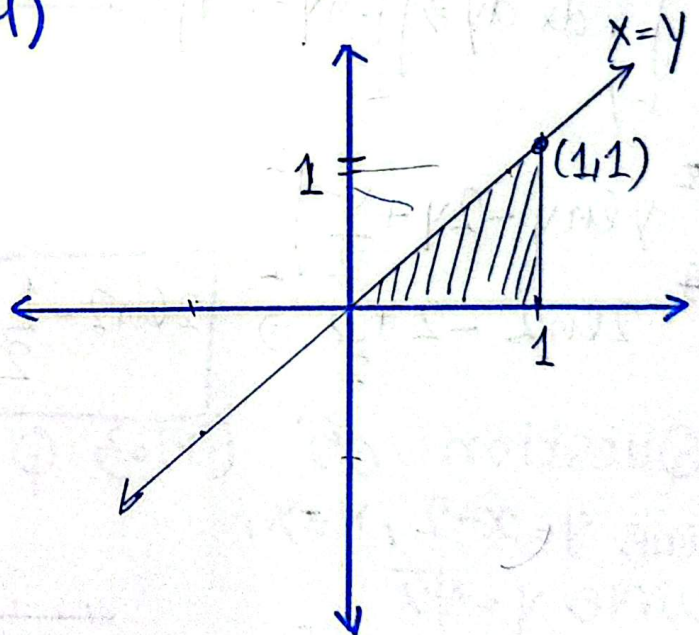
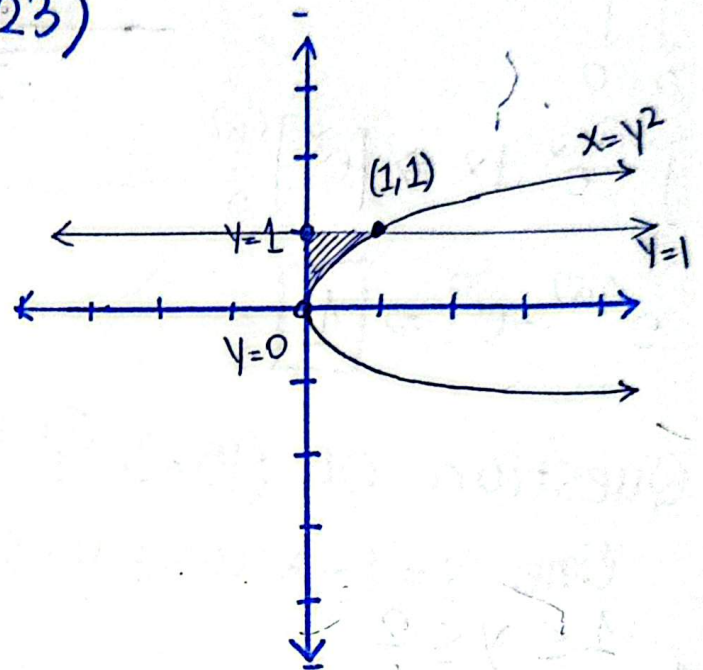
$$\int_0^1 \int_y^1 x^2 e^{xy} dx dy$$

$$\int_0^1 \int_0^x x^2 e^{xy} dy dx$$

$$\int_0^1 [x e^{xy}]_0^x dx$$

$$\int_0^1 (x e^{x^2} - x) dx$$

$$\left[\frac{1}{2} e^{x^2} - \frac{x^2}{2} \right]_0^1 = \boxed{\frac{e - 2}{2}}$$



Question 03 (Question 5-15.3)

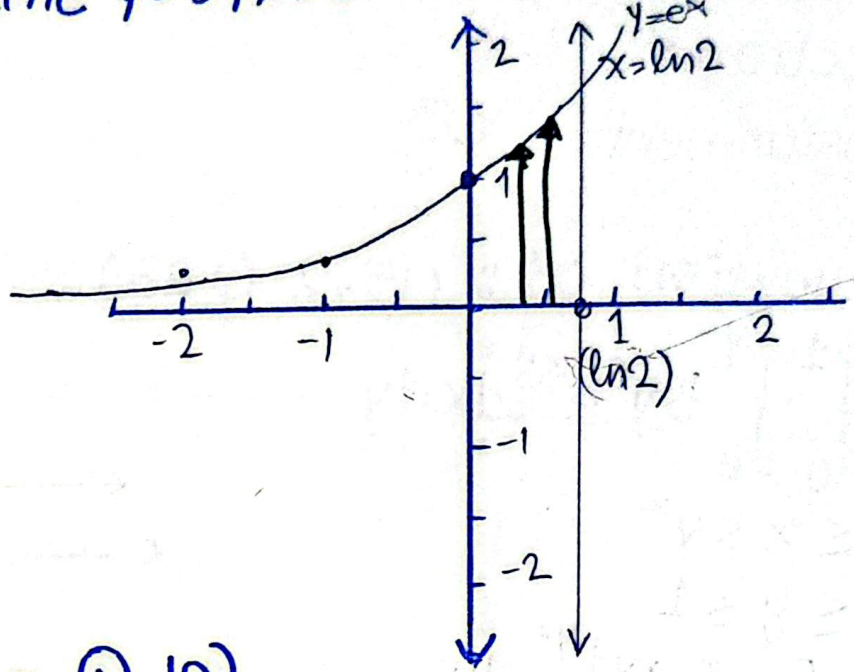
curve $y=e^x$ line $y=0, x=0$ and $x=\ln 2 \approx 0.69$
 $0 \leq y \leq e^x$

$$\ln 2 \quad e^x \quad 0 \leq x \leq \ln 2$$

$$\int_0^{\ln 2} \int_0^{e^x} dy dx$$

$$\int_0^{\ln 2} e^x dx \Rightarrow [e^x]_0^{\ln 2}$$

$$e^{\ln 2} - e^0 \Rightarrow \boxed{1}$$



Question 04 (15.3 Q 10)

line $y=1-x$ and $y=2$, curve $y=e^x$

$$1 \leq y \leq 2$$

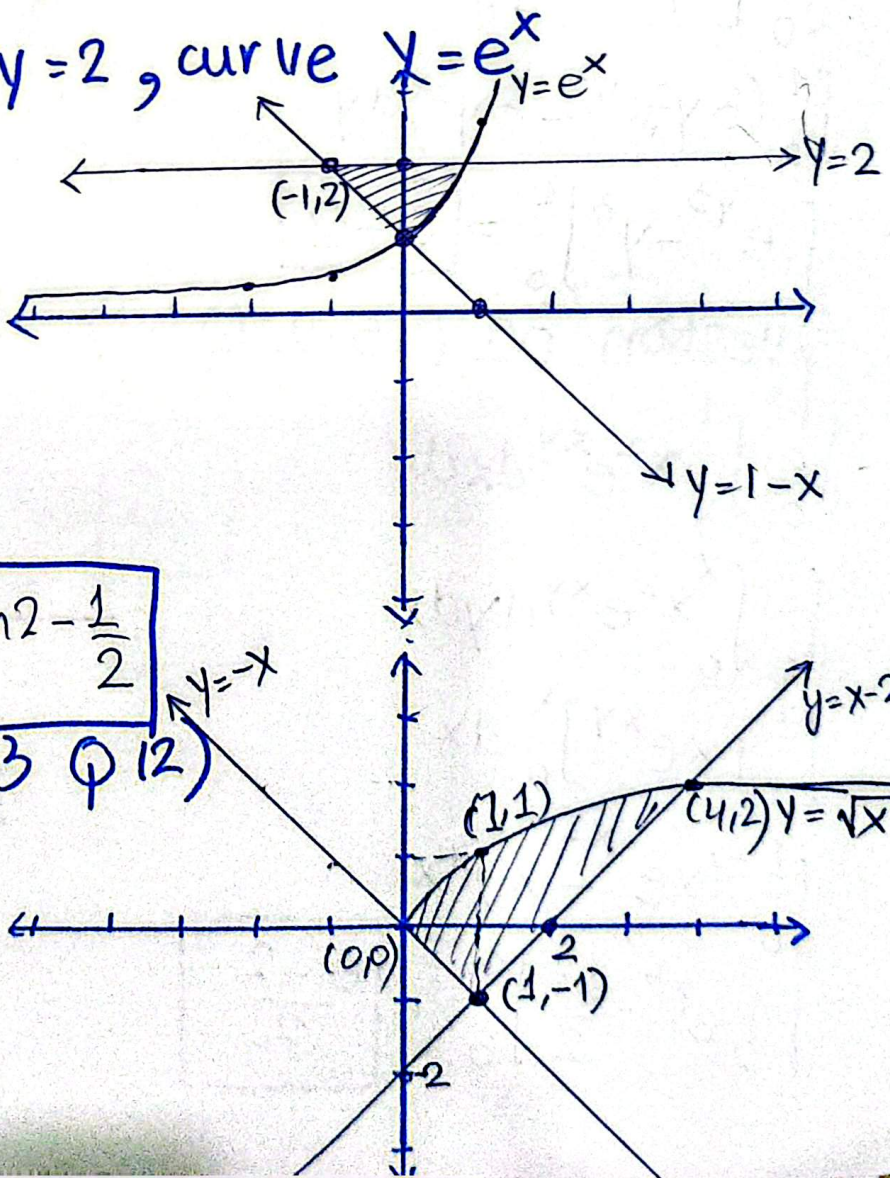
$$1-y \leq x \leq \ln y$$

$$\int_1^2 \int_{1-y}^{\ln y} dx dy \Rightarrow \int_1^2 [\ln y - 1 + y] dy$$

$$\left[y \ln y - 2y + \frac{y^2}{2} \right]_1^2$$

$$2 \ln 2 - 2 + \frac{3}{2} \Rightarrow$$

$$\boxed{2 \ln 2 - \frac{1}{2}}$$

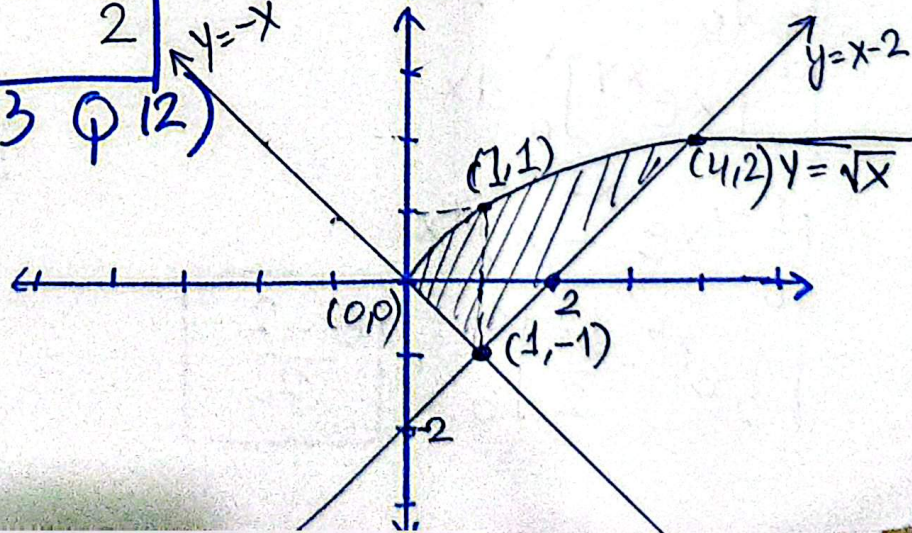


Question 05 (15.3 Q 12)

line $y=x-2, y=x,$

curve $y=\sqrt{x}$

$$\int_0^1 \int_{x-2}^{\sqrt{x}} dy dx + \int_1^4 \int_{x-2}^{\sqrt{x}} dy dx$$



$$\int_0^1 [y]_{-x}^{\sqrt{x}} dx + \int_1^4 [y]_{x-2}^{\sqrt{x}} dx$$

$$\int_0^1 \frac{2x^{3/2}}{3} + \frac{x^2}{2} dx + \int_1^4 \frac{2x^{3/2}}{3} - \frac{x^2}{2} + 2x$$

$$\frac{7}{6} + \frac{19}{6} \Rightarrow \boxed{\frac{13}{3}}$$

Question 06 (15.4 Q13)

$$\int_0^6 \int_0^y x dx dy$$

$$0 \leq x \leq y$$

For r :

$$y = 6$$

$$r \sin \theta = 6$$

$$r = \frac{6}{\sin \theta} \Rightarrow \boxed{r = 6 \csc \theta}$$

$$x = 0$$

$$r \cos \theta = 0$$

$$\boxed{r = 0}$$

$$\text{So, } \int_{\pi/4}^{\pi/2} \int_0^{6 \csc \theta} r \cos \theta \cdot r dr d\theta =$$

$$\int_{\pi/4}^{\pi/2} \cos \theta \int_0^{6 \csc \theta} \frac{r^3}{3} d\theta$$

$$\int_{\pi/4}^{\pi/2} \cos \theta [72 \csc^3 \theta] d\theta$$

$$72 \int_{\pi/4}^{\pi/2} \cos \theta \cdot \csc^3 \theta \cdot d\theta$$

$$-72 \int_{\pi/4}^{\pi/2} \frac{\cos \theta}{\sin \theta} \cdot \csc^2 \theta \cdot d\theta$$

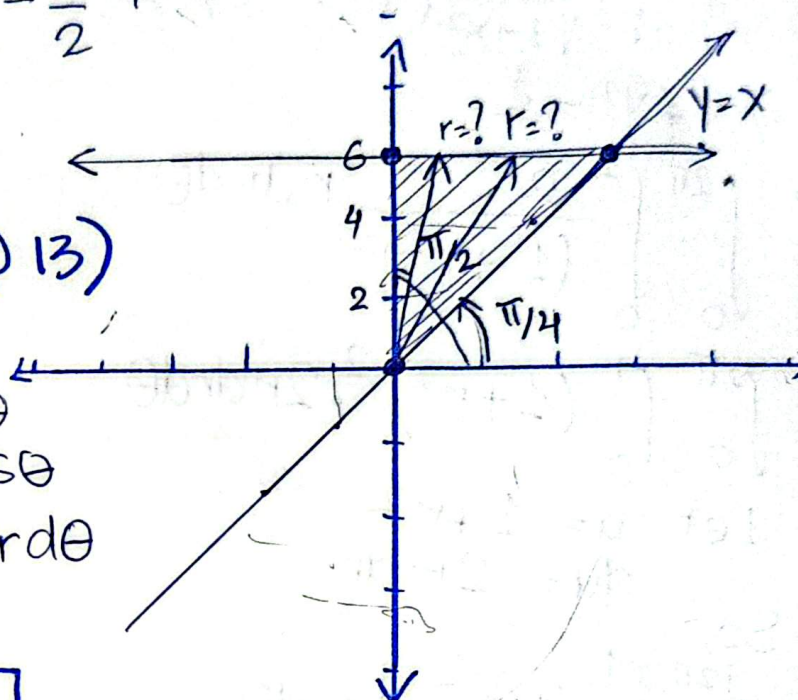
$$-72 \int_{\pi/4}^{\pi/2} \cot \theta \cdot (-\csc^2 \theta) d\theta$$

$$= -72 \left[\frac{\cot^2 \theta}{2} \right]_{\pi/4}^{\pi/2}$$

$$= -36 \left[\cot^2 \left(\frac{\pi}{2} \right) - \cot^2 \left(\frac{\pi}{4} \right) \right]$$

$$= -36 (0 - 1)$$

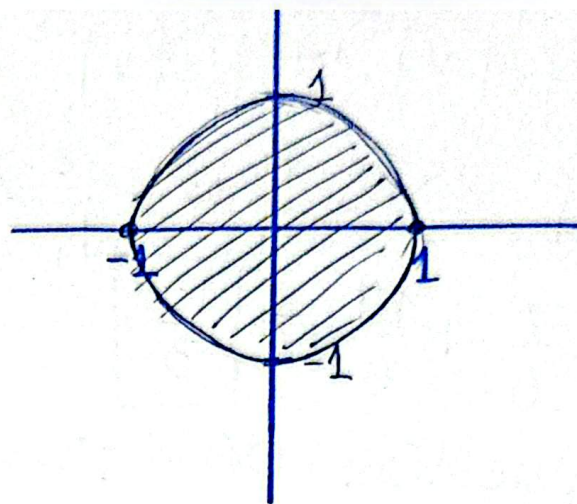
$$= \boxed{36}$$



Question 07 (15.4 Q 18)

$$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{2}{(1+x^2+y^2)^2} dy dx$$

$$\Rightarrow x^2 + y^2 = 1$$



$$\int_0^{2\pi} \int_0^1 \frac{2}{(1+r^2)^2} r dr d\theta$$

$$\int_0^{2\pi} \int_0^1 (1+r^2)^{-2} \cdot 2r dr d\theta$$

$$\text{Let } u = 1+r^2 \\ du = 2r dr$$

$$\text{So, } \int_0^{2\pi} \int_0^1 (u)^{-2} \cdot du \cdot d\theta$$

$$\int_0^{2\pi} \left[-\frac{1}{1+r^2} \right]_0^1 d\theta$$

$$\int_0^{2\pi} \frac{1}{2} d\theta \Rightarrow \frac{1}{2} [\theta]_0^{2\pi} \Rightarrow \frac{1}{2} (2\pi) \Rightarrow \boxed{\pi}$$

Question 08 (15.4 Q 22)

$$\int_1^2 \int_0^{\sqrt{2x-x^2}} \frac{1}{(x^2+y^2)^2} dy dx$$

$$y^2 = 2x - x^2$$

$$x^2 + y^2 - 2x = 0$$

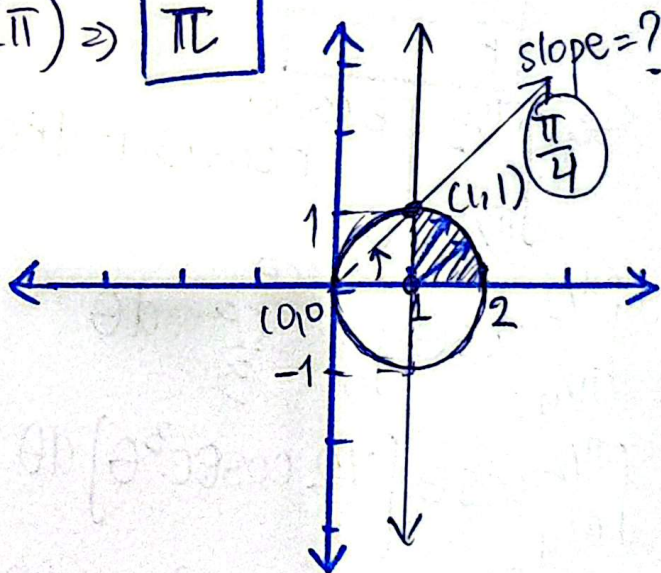
$$y^2 + x^2 - 2x = 0$$

$$y^2 + (x-1)^2 - 1 = 0$$

$$y^2 + (x-1)^2 = 1$$

$$(y-0)^2 + (x-1)^2 = 1 \quad \text{This circle's centre is}$$

$$\text{slope} \Rightarrow \tan \theta = \frac{1-0}{1-0} \Rightarrow \theta = \boxed{\frac{\pi}{4}} \quad (1,0)$$



For R :-

$$x^2 + y^2 - 2x = 0$$

$$r^2 - 2r \cos \theta = 0$$

$$\boxed{r = 2 \cos \theta}$$

$$x = 1$$

$$r \cos \theta = 1$$

$$r = \frac{1}{\cos \theta} \Rightarrow \boxed{r = \sec \theta}$$

$$= \frac{1}{2} \int_0^{\pi/4} \int_{\sec \theta}^{2 \cos \theta} (r^2)^{-2} \cdot 2r \, dr \, d\theta \Rightarrow \frac{1}{2} \int_0^{\pi/4} \int_{\sec \theta}^{2 \cos \theta} \frac{1}{r^2} \, d\theta$$

$$\frac{1}{2} \int_0^{\pi/4} \left[\cos^2 \theta - \frac{\sec^2 \theta}{4} \right] d\theta$$

$$\frac{1}{2} \int_0^{\pi/4} \left[\frac{1 + \cos 2\theta}{2} - \frac{\sec^2 \theta}{4} \right] d\theta \Rightarrow \frac{1}{2} \int_0^{\pi/4} \left(\frac{2 + 2 \cos 2\theta - \sec^2 \theta}{4} \right) d\theta$$

$$= \frac{1}{2} \cdot \frac{1}{4} \left[2\theta + \sin 2\theta - \tan \theta \right]_0^{\pi/4}$$

$$= \frac{1}{8} \left(\frac{\pi}{2} \right) + 1 - 1 \Rightarrow \boxed{\frac{\pi}{16}}$$

Question 09 (15.5 Q 18)

$$\int_0^1 \int_1^{\sqrt{e}} \int_1^e se^s \ln r \frac{(\ln t)^2}{t} dt dr ds$$

$$= \int_0^1 \int_1^{\sqrt{e}} (se^s \ln r) \left[\frac{1}{3} (\ln t)^3 \right]_1^e dr ds$$

$$= \int_0^1 \int_1^{\sqrt{e}} \frac{se^s}{3} \ln r \cdot dr ds = \int_0^1 \frac{se^s}{3} [r \ln r - r]_1^{\sqrt{e}} ds$$

$$= \frac{2 - \sqrt{e}}{6} \int_0^1 se^s ds = \frac{2 - \sqrt{e}}{6} [se^s - e^s]_0^1$$

$$= \boxed{\frac{2 - \sqrt{e}}{6}}$$

Question 10 (15.5 Q20)

$$\int_0^7 \int_0^2 \int_0^{\sqrt{4-q^2}} \frac{q}{r+1} dp dq dr$$

$$= \int_0^7 \int_0^2 \frac{q\sqrt{4-q^2}}{r+1} dq dr$$

$$= \int_0^7 \frac{1}{3(r+1)} \left[-(4-q^2)^{3/2} \right]_0^2 dr$$

$$= \frac{8}{3} \int_0^7 \frac{1}{r+1} dr = \frac{8 \ln 8}{3}$$

$$= \boxed{8 \ln 2}$$